

Hardik Chemburkar

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EDUCATION

University of Arizona

M.S. in Data Science | GPA: 3.88 / 4.0

Tucson, AZ, USA

Aug. 2024 – May 2026

Thakur College of Engineering & Technology (TCET)

B.Tech in Artificial Intelligence & Machine Learning | GPA: 9.47 / 10

Mumbai, India

Aug. 2020 – May 2024

EXPERIENCE

Artificial Intelligence Intern

Hydrotek Farm LLC

Jan. 2023 – Mar. 2024

United States (Remote)

- Collected, processed, and analyzed large-scale sensor and image datasets from hydroponic farms, performing **data validation and cleansing** to ensure **data quality and integrity** before downstream analysis; conducted **exploratory data analysis** to identify patterns in crop health indicators and translated findings into **actionable insights** for operations management.
- Built and evaluated **predictive models** in Python (TensorFlow, Scikit-learn) with iterative **feature engineering**, improving detection accuracy from **35% to 82%** and presenting results to **cross-functional stakeholders**.

Data Science & Business Analytics Intern

The Sparks Foundation

Dec. 2022 – Jan. 2023

Singapore (Remote)

- Delivered end-to-end analytics projects spanning regression, clustering, and EDA — built a student performance **predictive model** achieving **95% training / 88% test accuracy** with rigorous evaluation (RMSE, MAE, R^2) and iterative feature tuning.
- Designed **insight-driven visualisations and reports** across retail, sports, and geopolitical datasets, translating complex analytical findings into clear, **stakeholder-ready narratives** for non-technical audiences.

ACADEMIC PROJECTS

DHLitSearch — Large-Scale Literature Analysis Using NLP | MS Capstone | University of Arizona

Spring 2026

- Architected and implemented an end-to-end **NLP data pipeline** — engineering a checkpoint-based **OpenAlex API** ingestion system that collected **490,482 papers**, applied **DOI-based deduplication**, and generated **BioBERT embeddings** (768-dim) stored in a **10 GB ChromaDB vector database** — achieving **92%+ cosine similarity** and estimated **Precision@10 of 70–80%** on domain-specific retrieval benchmarks.
- Led cross-functional team coordination across a 3-member group under faculty mentorship, managing proposal iteration, milestone tracking, and GitHub documentation; the system integrates a **multi-backend RAG pipeline** (Vertex AI → Gemini → Ollama → BART fallback) with **BERTopic** topic modelling and a **Streamlit** web application — replacing weeks of manual literature screening with cited, grounded answers in seconds.

Sentiment Analysis – Statistical NLP | University of Arizona – INFO 557

Spring 2025

- Constructed an **end-to-end NLP pipeline**: text preprocessing, statistical **feature extraction**, and classification using **TF-IDF** and ensemble methods, applying **predictive modelling** to support **prescriptive analytics** on large text corpora.
- Evaluated model performance across precision, recall, and F1 metrics; documented findings in a reproducible, well-structured report demonstrating alignment between **analytical rigour** and clear communication of results.

Tucson Crime Data Analysis | University of Arizona – INFO 526

Fall 2024

- Performed spatial and temporal **EDA** on public safety data to identify crime concentration zones and recurring trends; developed **dashboards and visualisations** for non-technical stakeholders, demonstrating **data storytelling** and **business intelligence** competency.

TECHNICAL SKILLS

Data & Analytics: Exploratory Data Analysis, Statistical Analysis, Predictive & Prescriptive Analytics, Data Mining, Data Modelling, Data Validation & Cleansing, Business Intelligence, Data Storytelling

ML & AI: Supervised & Unsupervised Learning, NLP, Transformer Models (BioBERT, BART), Vector Embeddings, Semantic Search, RAG Pipelines, Feature Engineering, Model Evaluation, Topic Modelling (BERTopic)

Tools & Technologies: Python, R, SQL, TensorFlow, Scikit-learn, Pandas, NumPy, ChromaDB, Streamlit, **Power BI**, Tableau, MySQL, MongoDB, Docker, Git/GitHub, Jupyter, Linux/Ubuntu

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) 2025 – Certified Data Science Professional

Oct. 2025

Foundation of Sports Analytics – University of Michigan

Jun. 2025

Data Science & Business Analytics – The Sparks Foundation

Jan. 2023

Linux Spoken Tutorial – IIT Bombay

May 2021